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COMPUTATIONAL ARRAY MICROPROCESSOR SYSTEM USING NON-CONSECUTIVE DATA FORMATTING

Abstract

A microprocessor system comprises a computational array and a hardware data formatter. The computational array includes a plurality of computation units that each operates on a corresponding value addressed from memory. The values operated by the computation units are synchronously provided together to the computational array as a group of values to be processed in parallel. The hardware data formatter is configured to gather the group of values, wherein the group of values includes a first subset of values located consecutively in memory and a second subset of values located consecutively in memory. The first subset of values is not required to be located consecutively in the memory from the second subset of values.

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Background/Summary

CROSS REFERENCE TO OTHER APPLICATIONS

[0001] Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference under 37 CFR 1.57.

BACKGROUND OF THE INVENTION

[0002] Processing for machine learning and artificial intelligence typically requires performing mathematical operations on large sets of data and often involves solving multiple convolution layers. Applications of machine learning, such as self-driving and driver-assisted automobiles, often utilize array computational operations to calculate matrix and vector results. For example, array computational operations may be used to compute convolutional layers such as when performing image processing on captured sensor data. In many situations, a large amount of data is required to perform the necessary computational operations. Traditional implementations of these operations often require loading each element of a computational operation from a unique memory address. For a convolution operation, the process typically requires calculating an individual memory address for each element. Moreover, there is a potential to incur an additional delay from the latency involved in reading each data element from memory. These performance penalties are magnified when performing wide convolution operations that involve large input matrices and many matrix elements. Traditional solutions for performing computational operations, such as relying on multiple graphical processing unit (GPU) cores, utilize parallel processing to decrease the time spent computing. However, these solutions are limited in throughput in part due to the latency incurred by reading input data from memory. Therefore, there exists a need for a microprocessor system with increased throughput that performs array computational operations without the need to perform computationally and latency expensive operations for each of the individual elements of the input data.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Various embodiments of the invention are disclosed in the following detailed description and the accompanying drawings.

[0004] FIG. 1 is a block diagram illustrating an embodiment of a microprocessor system for performing machine learning processing.

[0005] FIG. 2 is a flow diagram illustrating an embodiment of a process for performing machine learning processing.

[0006] FIG. 3 is a flow diagram illustrating an embodiment of a process for performing machine learning processing.

[0007] FIG. 4 is a block diagram illustrating an embodiment of a computation unit of a computational array.

[0008] FIG. 5 is a block diagram illustrating an embodiment of a cache-enabled microprocessor system for performing machine learning processing.

[0009] FIG. 6 is a block diagram illustrating an embodiment of a hardware data formatter, cache, and memory components of a microprocessor system.

[0010] FIG. 7 is a flow diagram illustrating an embodiment of a process for performing machine learning processing.

[0011] FIG. 8 is a flow diagram illustrating an embodiment of a process for retrieving input operands for a computational array.

DETAILED DESCRIPTION

[0012] The invention can be implemented in numerous ways, including as a process; an apparatus; a system; a composition of matter; a computer program product embodied on a computer readable storage medium; and/or a processor, such as a processor configured to execute instructions stored on and/or provided by a memory coupled to the processor. In this specification, these implementations, or any other form that the invention may take, may be referred to as techniques. In general, the order of the steps of disclosed processes may be altered within the scope of the invention. Unless stated otherwise, a component such as a processor or a memory described as being configured to perform a task may be implemented as a general component that is temporarily configured to perform the task at a given time or a specific component that is manufactured to perform the task. As used herein, the term 'processor' refers to one or more devices, circuits, and/or processing cores configured to process data, such as computer program instructions.

[0013] A detailed description of one or more embodiments of the invention is provided below along with accompanying figures that illustrate the principles of the invention. The invention is described in connection with such embodiments, but the invention is not limited to any embodiment. The scope of the invention is limited only by the claims and the invention encompasses numerous alternatives, modifications and equivalents. Numerous specific details are set forth in the following description in order to provide a thorough understanding of the invention. These details are provided for the purpose of example and the invention may be practiced according to the claims without some or all of these specific details. For the purpose of clarity, technical material that is known in the technical fields related to the invention has not been described in detail so that the invention is not unnecessarily obscured.

[0014] A microprocessor system for performing high throughput array computational operations is disclosed. In some embodiments, a microprocessor system includes a computational array (e.g., matrix processor) in communication with a hardware data formatter for aligning the data to minimize data reads and the latency incurred by reading input data for processing. For example, a matrix processor allows a plurality of elements of a matrix and/or vector to be loaded and processed in parallel together. Thus, using data formatted by one or more hardware data formatters, a computational operation such as a convolution operation may be performed by the computational array.

[0015] One technique includes loading a large number of consecutive elements (e.g., consecutive in memory) of a matrix/vector together and performing operations on the consecutive elements in parallel using the matrix processor. By loading consecutive elements together, a single memory load and/or cache check for the entire group of elements can be performed-allowing the entire group of elements to be loaded using minimal processing resources. However, requiring the input elements of each processing iteration of the matrix processor to be consecutive elements could

potentially require the matrix processor to load a large number of matrix/vector elements that are to be not utilized. For example, performing a convolution operation using a stride greater than one requires access to matrix elements that are not consecutive. If parallel input elements to the matrix processor are required to be consecutive, each processing iteration of the matrix processor is unable to fully utilize every individual input element for workloads only requiring non-consecutive elements. An alternative technique is to not require every individual input element of the matrix processor be consecutive (e.g., every individual input element can be independently specified without regard to whether it is consecutive in memory to a previous input element). This technique incurs significant performance costs since each referenced element incurs the cost of determining its memory address and performing a cache check for the individual element with the potential of an even more expensive load from memory in the case of a cache miss.

[0016] In an embodiment of a disclosed microprocessor system, the group of input elements of a matrix processor are divided into a plurality of subsets, wherein elements within each subset are required be consecutive but the different subsets are not required to be consecutive. This allows the benefit of reduce resources required to load consecutive elements within each subset while providing the flexibility of loading non-consecutive elements across the different subsets. For example, a hardware data formatter loads multiple subsets of elements where the elements of each subset are located consecutively in memory. By loading the elements of each subset together, a memory address calculation and cache check is performed only with respect to the start and end elements of each subset. In the event of a cache miss, an entire subset of elements is loaded together from memory. Rather than incurring a memory lookup penalty on a per element basis as with the previous discussed technique, a cache check is minimized to two checks for each subset (the start and end elements) and a single memory read for the entire subset in the event of a cache miss. Computational operations on non-consecutive elements, such as the performing convolution using a stride greater than one, are more efficient since the memory locations of the subsets need not be consecutively located in memory. Using the disclosed system and techniques, computational operations may be performed on non-consecutive elements with increased throughput and a high clock frequency.

[0017] In various embodiments, a computational array performs matrix operations involving input vectors and includes a plurality of computation units to receive M operands and N operands from the input vectors. Using a sequence of input vectors, a computational array may perform matrix operations such as a matrix multiplication. In some embodiments, the computation units are sub-circuits that include an arithmetic logic unit, an accumulator, a shadow register, and a shifter for performing operations such as generating dot-products and various processing for convolution. Unlike conventional graphical processing unit (GPU) or central processing unit (CPU) processing cores, where each core is configured to receive its own unique processing instruction, the computation units of the computational array each perform the same computation in parallel in response to an individual instruction received by the computational array.

[0018] In various embodiments, the data input to the computational array is prepared using a hardware data formatter. For example, a hardware data formatter is utilized to load and align data elements using subsets of elements where the elements of each subset are located consecutively in memory and the subsets need not be located consecutively in memory. In various embodiments, the various subsets may each have a memory location independent from other subsets. For example, the different subsets may be located non-consecutively in memory from one another. By restricting the data elements within a subset to consecutive data, multiple consecutive data elements are processed together, which minimizes the calculations and delay incurred when preparing the data for a computational array. For example, a subset of data elements may be cached as a consecutive sequence of data elements by performing a cache check on the start and end element and, in the event of a cache miss on either element, a single data read to load the entire subset from memory into a memory cache. Once all the data elements are available, the data may be provided together to

the computational array as a group of values to be processed in parallel.

[0019] In some embodiments, a microprocessor system comprises a computational array and a hardware data formatter. For example, a microprocessor system includes a matrix processor capable of performing matrix and vector operations. In various embodiments, the computational array includes a plurality of computation units. For example, the computation units may be sub-circuits of a matrix processor that include the functionality for performing one or more multiply, add, accumulate, and shift operations. As another example, computation units may be sub-circuits that include the functionality for performing a dot-product operation. In various embodiments, the computational array includes a sufficient number of computation units for performing multiple operations on the data inputs in parallel. For example, a computational array configured to receive M operands and N operands may include at least $M \times N$ computation units. In various embodiments, each of the plurality of computation units operates on a corresponding value formatted by a hardware data formatter and the values operated by the plurality of computation units are synchronously provided together to the computational array as a group of values to be processed in parallel. For example, values corresponding to elements of a matrix are processed by one or more hardware data formatters and provided to the computational array together as a group of values to be processing in parallel.

[0020] In various embodiments, a hardware data formatter is configured to gather the group of values to be processed in parallel by the computational array. For example, a hardware data formatter retrieves the values from memory, such as static random access memory (SRAM), via a cache. In some embodiments, in the event of a cache miss, the hardware data formatter loads the values into the cache from memory and subsequently retrieves the values from the cache. In various embodiments, the values provided to the computational array correspond to computational operands. For example, a hardware formatter may process M operands as an input vector to a computational array. In various embodiments, a second hardware formatter may process N operands as a second input vector to the computational array. In some embodiments, each hardware data formatter processes a group of values synchronously provided together to the computational array, where each group of values includes a first subset of values located consecutively in memory and a second subset of values located consecutively in memory, yet the first subset of values are not located consecutively in the memory from the second subset of values. For example, a hardware data formatter loads a first subset of values stored consecutively in memory and a second subset of values also stored consecutively in memory but with a gap in memory between the two subsets of values. Each subset of values is loaded as consecutive values into the hardware data formatter. To prepare an entire vector of inputs for a computational array, the hardware data formatter performs loads based on the number of subsets instead of based on the total number of elements needed for an input operand to a computational array.

[0021] FIG. 1 is a block diagram illustrating an embodiment of a microprocessor system for performing machine learning processing. In the example shown, microprocessor system **100** includes control unit **101**, data input **103**, data formatter **104**, weight input **105**, weight formatter **106**, matrix processor **107**, vector engine **111**, and post-processing unit **115**. Data input **103** and weight input **105** are input data that is fed to hardware data formatters data formatter **104** and weight formatter **106**. In some embodiments, data input **103** and/or weight input **105** are retrieved from a memory (not shown), which may include a memory cache or buffer to reduce latency when reading data. In the example shown, data formatter **104** and weight formatter **106** are hardware data formatters for preparing data for matrix processor **107**. In some embodiments, data formatter **104** and weight formatter **106** include a logic circuit for preparing data for matrix processor **107** and/or a memory cache or buffer for storing and processing input data. For example, data formatter **104** may prepare N operands from a two-dimensional array retrieved from data input **103** that correspond to image data. Weight formatter **106** may prepare M operands retrieved from weight input **105** that correspond to a vector of weight values. Data formatter **104** and weight formatter

106 prepare the N and M operands to be processed by matrix processor **107**. In some embodiments, microprocessor system **100**, including at least hardware data formatters data formatter **104** and weight formatter **106**, matrix processor **107**, vector engine **111**, and post-processing unit **115**, perform the processes described below with respect to FIGS. 2, 3, 7, and 8.

[0022] In some embodiments, matrix processor **107** is a computational array that includes a plurality of computation units. For example, a matrix processor receiving M operands and N operands from weight formatter **106** and data formatter **104**, respectively, includes $M \times N$ computation units. In the figure shown, the small squares inside matrix processor **107** depict that matrix processor **107** includes a logical two-dimensional array of computation units. Computation unit **109** is one of a plurality of computation units of matrix processor **107**. In some embodiments, each computation unit is configured to receive one operand from data formatter **104** and one operand from weight formatter **106**. In some embodiments, the computation units are configured according to a logical two-dimensional array but the matrix processor is not necessarily fabricated with computation units laid out as a physical two-dimensional array. For example, the i-th operand of data formatter **104** and the j-th operand of weight formatter **106** are configured to be processed by the i-th $\times$ j-th computation unit of matrix processor **107**.

[0023] In various embodiments, the data width of components data formatter **104**, weight formatter **106**, matrix processor **107**, vector engine **111**, and post-processing unit **115** are wide data widths and include the ability to transfer more than one operand in parallel. In some embodiments, data formatter **104** and weight formatter **106** are each 96-bytes wide. In some embodiments, data formatter **104** is 192-bytes wide and weight formatter **106** is 96-bytes wide. In various embodiments, the width of data formatter **104** and weight formatter **106** is dynamically configurable. For example, data formatter **104** may be dynamically configured to 96 or 192 bytes and weight formatter **106** may be dynamically configured to 96 or 48 bytes. In some embodiments, the dynamic configuration is controlled by control unit **101**. In various embodiments, a data width of 96 bytes allows 96 operands to be processed in parallel. For example, in an embodiment with data formatter **104** configured to be 96-bytes wide, data formatter **104** can transfer 96 operands to matrix processor **107** in parallel.

[0024] In various embodiments, data input **103** and weight input **105** are input data to corresponding hardware data formatters data formatter **104** and weight formatter **106** based on memory addresses calculated by the hardware data formatters. In some embodiments, data formatter **104** and/or weight formatter **106** retrieves via data input **103** and weight input **105**, respectively, a stream of data corresponding to one or more subsets of values stored consecutively in memory. Data formatter **104** and/or weight formatter **106** may retrieve one or more subsets of values stored consecutively in memory and prepare the data as input values for matrix processor **107**. In various embodiments, the one or more subsets of values are not themselves stored consecutively in memory with other subsets of values. In some embodiments, data input **103** and/or weight input **105** are retrieved from memory (not shown in FIG. 1) that contains a single read port. In some embodiments, the memory contains a limited number of read ports and the number of read ports is fewer than the data width of components data formatter **104**, weight formatter **106**, matrix processor **107**, vector engine **111**, and/or post-processing unit **115**. In some embodiments, the memory includes a cache and a hardware data formatter, such as data formatter **104** and weight formatter **106**, which will perform a cache check to determine whether each subset of values is in the cache prior to issuing a read request to memory. In the event the subset of values is cached, a hardware data formatter (e.g., data formatter **104** or weight formatter **106**) will retrieve the data from the cache. In various embodiments, in the event of a cache miss, the hardware data formatter (e.g., data formatter **104** or weight formatter **106**) will retrieve the entire subset of values from memory and populate the cache with the retrieved values.

[0025] In various embodiments, matrix processor **107** is configured to receive N bytes from data formatter **104** and M bytes from weight formatter **106** and includes at least $M \times N$ computation

units. For example, matrix processor **107** may be configured to receive 96 bytes from data formatter **104** and 96 bytes from weight formatter **106** and includes at least 96×96 computation units. As another example, matrix processor **107** may be configured to receive 192 bytes from data formatter **104** and 48 bytes from weight formatter **106** and includes at least 192×48 computation units. In various embodiments, the dimensions of matrix processor **107** may be dynamically configured. For example, the default dimensions of matrix processor **107** may be configured to receive 96 bytes from data formatter **104** and 96 bytes from weight formatter **106** but the input dimensions may be dynamically configured to 192 bytes and 48 bytes, respectively. In various embodiments, the output size of each computation unit is equal to or larger than the input size. For example, in some embodiments, the input to each computation unit is two 1-byte operands, one corresponding to an operand from data formatter **104** and one from weight formatter **106**, and the output of processing the two operands is a 4-byte result. As another example, matrix processor **107** may be configured to receive 96 bytes from data formatter **104** and 96 bytes from weight formatter **106** and output 96 4-byte results. In some embodiments, the output of matrix processor **107** is a vector. For example, a matrix processor configured to receive two 96-wide input vectors, where each element (or operand) of the input vector is one byte in size, can output a 96-wide vector result where each element of the vector result is 4-bytes in size.

[0026] In various embodiments, each computation unit of matrix processor **107** is a sub-circuit that includes an arithmetic logic unit, an accumulator, and a shadow register. In the example shown, the computation units of matrix processor **107** can perform an arithmetic operation on the M operands and N operands from weight formatter **106** and data formatter **104**, respectively. In various embodiments, each computation unit is configured to perform one or more multiply, add, accumulate, and/or shift operations. In some embodiments, each computation unit is configured to perform a dot-product operation. For example, in some embodiments, a computation unit may perform multiple dot-product component operations to calculate a dot-product result. For example, the array of computation units of matrix processor **107** may be utilized to perform convolution steps required for performing inference using a machine learning model. A two-dimensional data set, such as an image, may be formatted and fed into matrix processor **107** using data formatter **104** and data input **103**, one vector at a time. In parallel, a filter of weights may be applied to the two-dimensional data set by formatting the weights and feeding them as a vector into matrix processor **107** using weight formatter **106** and weight input **105**. Corresponding computation units of matrix processor **107** perform a matrix processor instruction on the corresponding operands of the weight and data inputs in parallel.

[0027] In some embodiments, vector engine **111** is a vector computational unit that is communicatively coupled to matrix processor **107**. Vector engine **111** includes a plurality of processing elements including processing element **113**. In the figure shown, the small squares inside vector engine **111** depict that vector engine **111** includes a plurality of processing elements arranged as a vector. In some embodiments, the processing elements are arranged in a vector in the same direction as data formatter **104**. In some embodiments, the processing elements are arranged in a vector in the same direction as weight formatter **106**. In various embodiments, the data size of the processing elements of vector engine **111** is the same size or larger than the data size of the computation units of matrix processor **107**. For example, in some embodiments, computation unit **109** receives two operands each 1 byte in size and outputs a result 4 bytes in size. Processing element **113** receives the 4-byte result from computation unit **109** as an input 4 bytes in size. In various embodiments, the output of vector engine **111** is the same size as the input to vector engine **111**. In some embodiments, the output of vector engine **111** is smaller in size compared to the input to vector engine **111**. For example, vector engine **111** may receive up to 96 elements each 4 bytes in size and output 96 elements each 1 byte in size. As described above, in some embodiments, the communication channel from data formatter **104** and weight formatter **106** to matrix processor **107** is 96-elements wide with each element 1 byte in size and matches the output size of vector engine

111 (96-elements wide with each element 1 byte in size).

[0028] In some embodiments, the processing elements of vector engine **111**, including processing element **113**, each include an arithmetic logic unit (ALU) (not shown). For example, in some embodiments, the ALU of each processing element is capable of performing arithmetic operations. In some embodiments, each ALU of the processing elements is capable of performing in parallel a rectified linear unit (ReLU) function and/or scaling functions. In some embodiments, each ALU is capable of performing a non-linear function including non-linear activation functions. In various embodiments, each processing element of vector engine **111** includes one or more flip-flops for receiving input operands. In some embodiments, each processing element has access to a slice of a vector engine accumulator and/or vector registers of vector engine **111**. For example, a vector engine capable of receiving 96-elements includes a 96-element wide accumulator and one or more 96-element vector registers. Each processing element has access to a one-element slice of the accumulator and/or vector registers. In some embodiments, each element is 4-bytes in size. In various embodiments, the accumulator and/or vector registers are sized to fit at least the size of an input data vector. In some embodiments, vector engine **111** includes additional vector registers sized to fit the output of vector engine **111**.

[0029] In some embodiments, the processing elements of vector engine **111** are configured to receive data from matrix processor **107** and each of the processing elements can process the received portion of data in parallel. As one example of a processing element, processing element **113** of vector engine **111** receives data from computation unit **109** of matrix processor **107**. In various embodiments, vector engine **111** receives a single vector processor instruction and in turn each of the processing elements performs the processor instruction in parallel with the other processing elements. In some embodiments, the processor instruction includes one or more component instructions, such as a load, a store, and/or an arithmetic logic unit operation. In various embodiments, a no-op operation may be used to replace a component instruction.

[0030] In the example shown, the dotted arrows between data formatter **104** and matrix processor **107**, weight formatter **106** and matrix processor **107**, matrix processor **107** and vector engine **111**, and vector engine **111** and post-processing unit **115** depict couplings between the respective pairs of components that are capable of sending multiple data elements such as a vector of data elements. As an example, the communication channel between matrix processor **107** and vector engine **111** may be 96×32 bits wide and support transferring 96 elements in parallel where each element is 32 bits in size. As another example, the communication channel between vector engine **111** and post-processing unit **115** may be 96×1 byte wide and support transferring 96 elements in parallel where each element is 1 byte in size. In various embodiments, data input **103** and weight input **105** are retrieved from a memory module (not shown in FIG. 1). In some embodiments, vector engine **111** is additionally coupled to a memory module (not shown in FIG. 1) and may receive input data from the memory module in addition or alternatively to input from matrix processor **107**. In the various embodiments, a memory module is typically a static random access memory (SRAM).

[0031] In some embodiments, one or more computation units of matrix processor **107** may be grouped together into a lane such that matrix processor **107** has multiple lanes. In various embodiments, the lanes of matrix processor **107** may be aligned with either data formatter **104** or weight formatter **106**. For example, a lane aligned with weight formatter **106** includes a set of computation units that are configured to receive as input every operand of weight formatter **106**. Similarly, a lane aligned with data formatter **104** includes a set of computation units that are configured to receive as input every operand of data formatter **104**. In the example shown in FIG. 1, the lanes are aligned along weight formatter **106** in a vertical column and each lane feeds to a corresponding lane of vector engine **111**. In some embodiments, each lane is a vertical column of sub-circuits that include multiply, add and/or accumulate, and shift functionality. In some embodiments, matrix processor **107** includes a matrix of tiles and each tile is a matrix of computation units. For example, a 96×96 matrix processor may include a matrix of 6×6 tiles, where

each tile includes 16×16 computation units. In some embodiments, a vertical lane is a single column of tiles. In some embodiments, a horizontal lane is a single row of tiles. In various embodiments, the dimensions of the lane may be configured dynamically and may be utilized for performing alignment operations on the input to matrix processor **107**, vector engine **111**, and/or post-processing unit **115**. In some embodiments, the dynamic configuration is performed by or using control unit **101** and/or with using processor instructions and/or control signals controlled by control unit **101**.

[0032] In some embodiments, control unit **101** synchronizes the processing performed by data formatter **104**, weight formatter **106**, matrix processor **107**, vector engine **111**, and post-processing unit **115**. For example, control unit **101** may send processor specific control signals and/or instructions to each of data formatter **104**, weight formatter **106**, matrix processor **107**, vector engine **111**, and post-processing unit **115**. In some embodiments, a control signal is utilized instead of a processor instruction. Control unit **101** may send matrix processor instructions to matrix processor **107**. A matrix processor instruction may be a computational array instruction that instructs a computational array to perform an arithmetic operation, such as a dot-product or dot-product component, using specified operands from data input **103** and/or weight input **105** that are formatted by data formatter **104** and/or weight formatter **106**, respectively. Control unit **101** may send vector processor instructions to vector engine **111**. For example, a vector processor instruction may include a single processor instruction with a plurality of component instructions to be executed together by the vector computational unit. Control unit **101** may send post-processing instructions to post-processing unit **115**. In various embodiments, control unit **101** synchronizes data that is fed to matrix processor **107** from data formatter **104** and weight formatter **106**, to vector engine **111** from matrix processor **107**, and to post-processing unit **115** from vector engine **111**. In some embodiments, control unit **101** synchronizes the data between different components of microprocessor system **100** including between data formatter **104**, weight formatter **106**, matrix processor **107**, vector engine **111**, and/or post-processing unit **115** by utilizing processor specific memory, queue, and/or dequeue operations and/or control signals. In some embodiments, data and instruction synchronization is performed by control unit **101**. In some embodiments, data and instruction synchronization is performed by control unit **101** that includes one or more sequencers to synchronize processing between data formatter **104**, weight formatter **106**, matrix processor **107**, vector engine **111**, and/or post-processing unit **115**.

[0033] In some embodiments, data input **103**, data formatter **104**, weight input **105**, weight formatter **106**, matrix processor **107**, and vector engine **111** are utilized for processing convolution layers. For example, matrix processor **107** may be used to perform calculations associated with one or more convolution layers of a convolution neural network. Data formatter **104** and weight formatter **106** may be utilized to prepare matrix and/or vector data in a format for processing by matrix processor **107**. Data input **103** may include image data such as one or more image channels captured by sensors (not shown), where sensors include, as an example, cameras mounted to a vehicle. Weight input **105** may include weights determined by training a machine learning model for autonomous driving. In some embodiments, vector engine **111** is utilized for performing non-linear functions such as an activation function on the output of matrix processor **107**. For example, matrix processor **107** may be used to calculate a dot-product and vector engine **111** may be used to perform an activation function such as a rectified linear unit (ReLU) or sigmoid function. In some embodiments, post-processing unit **115** is utilized for performing pooling operations. In some embodiments, post-processing unit **115** is utilized for formatting and storing the processed data to memory and may be utilized for synchronizing memory writing latency.

[0034] FIG. 2 is a flow diagram illustrating an embodiment of a process for performing machine learning processing. In some embodiments, the process of FIG. 2 is utilized to implement a convolutional neural network using sensor input data such as images and learned weights. In various embodiments, the process of FIG. 2 may be repeated for multiple convolution layers by

using the output of the process of FIG. 2 as the input for the next convolution layer. In some embodiments, the processing is performed in the context of self-driving or driver-assisted vehicles to identify objects in a scene such as street signs, vehicles, pedestrians, and lane markers, among other objects. Other sensor data, including non-image sensor data, such as ultrasonic, radar, and LiDAR, may also be utilized as input data. In various embodiments, the process of FIG. 2 utilizes a microprocessor system such as is microprocessor system **100** of FIG. **1**.

[0035] At **201**, input channels are received as input data to the microprocessor system. For example, vision data is captured using sensors and may include one or more channels corresponding to different color channels for the colors red, green, and blue. In various embodiments, multiple channels may be utilized as the different channels may contain different forms of information. As another example, non-sensor data may be utilized as input data. In various embodiments, the input channels may be loaded from memory via a cache using subsets of consecutively stored data in memory. In some embodiments, the input channels may be retrieved and/or formatted for processing using a hardware data formatter such as data formatter **104** of FIG. **1**.

[0036] At **203**, one or more filters are received for processing the input channels. For example, a filter in the form of a matrix contains learned weights and is used to identify activations in the channels. In some embodiments, the filter is a square matrix kernel smaller than the input channel. In various embodiments, filters may be utilized to identify particular shapes, edges, lines, and other features and/or activations in the input data. In some embodiments, the filters and associated weights that make up the filter are created by training a machine learning model using a training corpus of data similar to the input data. In various embodiments, the received filters may be streamed from memory. In some embodiments, the filters may be retrieved and/or formatted for processing using a hardware data formatter such as weight formatter **106** of FIG. **1**.

[0037] At **205**, one or more feature layers are determined using the received input channels and filters. In various embodiments, the feature layers are determined by performing one or more convolution operations using a computational array such as matrix processor **107** of FIG. **1**. In some embodiments, the one or more output feature layers are determined by repeatedly performing a dot-product between different small regions of an input channel and the weights of the filter. In various embodiments, each filter is used to create a single feature layer by performing a two-dimensional convolution using the filter. In some embodiments, the input data is padded to adjust for the size of the output feature layer. In various embodiments, a stride parameter is utilized and may impact the size of the output feature layer. In various embodiments, a bias parameter may be utilized. For example, a bias term may be added to the resulting values of convolution for each element of a feature layer.

[0038] At **207**, an activation function is performed on one or more feature layers. For example, an element-wise activation function, such as a rectified linear unit (ReLU) function, is performed using a vector processor such as vector engine **111** of FIG. **1** to create an activation layer. In various embodiments, different activation functions, such as a non-linear activation function, including ReLU and sigmoid, may be utilized to create an activation layer for each feature layer.

[0039] At **209**, pooling is performed on the activation layers created at **207**. For example, a pooling layer is generated by a post-processing unit such as post-processing unit **115** of FIG. **1** using the activation layer generated at **207**. In some embodiments, the pooling layer is generated to down sample the activation layer. In various embodiments, different filter sizes may be utilized to create a pooling layer based on the desired output size. In various embodiments, different pooling techniques, such as maxpooling, are utilized. In various embodiments, pooling parameters include kernel size, stride, and/or spatial extent, among others. In some embodiments, the pooling layer is an optional layer and may be implemented when appropriate.

[0040] In various embodiments, the process of FIG. 2 is utilized for each layer of a convolution neural network (CNN). Multiple passes of the process of FIG. 2 may be utilized to implement a

multi-layer CNN. For example, the output of **209** may be utilized as input channels at **201** to calculate output layers of an intermediate layer. In some embodiments, a CNN is connected to one or more additional non-CNN layers for classification, object detection, object segmentation, and/or other appropriate goals. In some embodiments, the additional non-CNN layers are implemented using a microprocessor system such as is microprocessor system **100** of FIG. **1**.

[0041] FIG. **3** is a flow diagram illustrating an embodiment of a process for performing machine learning processing. In some embodiments, the process of FIG. **3** is utilized to perform inference on sensor data by performing computational operations, such as convolution operations, and element-wise activation functions. In some embodiments, the process of FIG. **3** is performed using a microprocessor system such as is microprocessor system **100** of FIG. **1**. In various embodiments, steps **301** and **303** are performed at **201** of FIG. **2** using data input **103** and data formatter **104** of FIG. **1**, steps **305** and **307** are performed at **203** of FIG. **2** using weight input **105** and weight formatter **106** of FIG. **1**, step **309** is performed at **205** of FIG. **2** using matrix processor **107** of FIG. **1**, step **311** is performed at **207** of FIG. **2** using vector engine **111** of FIG. **1**, and step **313** is performed at **209** of FIG. **2** using post-processing unit **115** of FIG. **1**.

[0042] At **301**, data input is received. For example, data input corresponding to sensor data is received by a hardware data formatter for formatting. In some embodiments, data input is data input **103** of FIG. **1** and is received by data formatter **104** of FIG. **1**. In various embodiments, a hardware data formatter requests the data input from memory as read requests based on subsets of values stored consecutively in memory. For example, a hardware data formatter may first check a cache of the memory for the requested data values and in the event of a cache miss, the read request will retrieve the data values from memory. In various embodiments, checking for a cache hit or miss requires calculating the start address and end address of the subset of requested data values. In some embodiments, a data request populates the cache with the requested values along with additional data to fill a cache line. In some embodiments, the data is streamed in from memory and may bypass the cache.

[0043] At **303**, data input is formatted using a hardware data formatter. For example, a hardware data formatter such as data formatter **104** of FIG. **1** formats the received data input for processing by a computational array such as matrix processor **107** of FIG. **1**. The hardware data formatter may format the received data input into an input vector of operands for a computational array. In some embodiments, the hardware data formatter further performed the requesting of data received at **301**. In some embodiments, the hardware data formatter will format at least one of the operands of a convolution operation. For example, each two-dimensional region corresponding to an input channel of vision data for a convolution operation involving a filter will be formatted by the hardware data formatter into a vector operand for the computational array. The vectors corresponding to the regions are grouped together by their n-th elements and fed to the computation array at a rate of at most one element from each vector per clock cycle. In some embodiments, the hardware data formatter will select the appropriate elements for performing convolution of a filter with the data input by formatting each region of the data input into a vector and feeding each element of the appropriate vector to a corresponding computation unit of a computational array. In some embodiments, a bias parameter is introduced using the hardware data formatter.

[0044] At **305**, weight input is received. For example, weight input corresponding to machine learning weights of a filter are received by a hardware data formatter for formatting. In some embodiments, weight input is weight input **105** of FIG. **1** and is received by weight formatter **106** of FIG. **1**. In various embodiments, a hardware data formatter requests the weight input from memory as read requests based on subsets of values stored consecutively in memory. For example, a hardware data formatter may first check a cache of the memory for the requested weight values and in the event of a cache miss, the read request will retrieve the weight values from memory. In various embodiments, checking for a cache hit or miss requires calculating the start address and end address of the subset of requested weight values. In some embodiments, a weight data request

populates the cache with the requested weight values. In some embodiments, the data for weights is streamed in from memory and may bypass the cache. In some embodiments, the weight input includes a bias parameter.

[0045] At **307**, weight input is formatted using a hardware data formatter. For example, a hardware data formatter such as weight formatter **106** of FIG. **1** formats the received weight input for processing by a computational array such as matrix processor **107** of FIG. **1**. The hardware data formatter may format the received weight input into an input vector of operands for a computational array. In some embodiments, the hardware data formatter further performed the requesting of data received at **305**. In some embodiments, the hardware data formatter will format at least one of the operands of a convolution operation. For example, a filter for a convolution operation will be formatted by the hardware data formatter into a vector operand for the computational array. In some embodiments, the hardware data formatter will select the appropriate elements for performing convolution of a filter with the data input by formatting the filter into a vector and feeding each element of the vector to a corresponding computation unit of a computational array. In some embodiments, a bias parameter is introduced using the hardware data formatter.

[0046] At **309**, matrix processing is performed. For example, the operands formatted at **303** and **307** are received by each of the computation units of a computational array for processing. In some embodiments, the matrix processing is performed using a matrix processor such as matrix processor **107** of FIG. **1**. In some embodiments, a dot-product is performed at each appropriate computation unit of the computational array using respective vectors received by hardware data formatters such as data formatter **104** and weight formatter **106** of FIG. **1**. In some embodiments, only a subset of the matrix processor's computation units is utilized. For example, a computational array with 96×96 computation units may utilize only 64×64 computation units in the event the data input is 64 vectors and the weight input is 64 vectors. In various embodiments, the number of computation units utilized is based on the size on the data input and/or weight input. In some embodiments, the computation units each perform one or more of multiply, add, accumulate, and/or shift operations. In some embodiments, the computation units each perform one or more of multiply, add, accumulate, and/or shift operations each clock cycle. In some embodiments, a bias parameter is received and added to the calculated dot-product as part of the matrix processing performed.

[0047] At **311**, vector processing is performed. For example, an element-wise activation function may be performed on the result of the matrix processing performed at **309**. In some embodiments, an activation function is a non-linear activation function such as a rectified linear unit (ReLU), sigmoid, or other appropriate function. In some embodiments, the vector processor is utilized to implement scaling, normalization, or other appropriate techniques. For example, a bias parameter may be introduced to the result of a dot-product using the vector processor. In some embodiments, the result of **311** is a series of activation maps or activation layers. In some embodiments, vector processing is performed using a vector engine such as vector engine **111** of FIG. **1**.

[0048] At **313**, post-processing is performed. For example, a pooling layer may be implemented using a post-processing processor such as post-processing unit **115** of FIG. **1**. In various embodiments, different post-processing techniques, including different pooling techniques such as maxpooling, may be implemented during the post-processing stage of **313**.

[0049] In various embodiments, the process of FIG. **3** is utilized for each layer of a convolution neural network (CNN). Multiple passes of the process of FIG. **3** may be utilized to implement a multi-layer CNN. For example, the output of **313** may be utilized as data input for step **301**. In some embodiments, the process of FIG. **3** must be repeated one or more times to complete a single layer. For example, in the scenario where the sensor data is larger in dimension than the number of computation units of the computational array, the sensor data may be sliced into smaller regions that fit the computational array and the process of FIG. **3** is repeated on each of the sliced regions.

[0050] FIG. 4 is a block diagram illustrating an embodiment of a computation unit of a computational array. In the example shown, computation unit **400** includes input values weight **402**, data **404**, and ResultIn **406**; signals ClearAcc signal **408**, Clock signal **410**, ResultEnable signal **412**, ResultCapture signal **414**, and ShiftEn signal **416**; components accumulator **424**, multiplexer **426**, shadow register **428**, multiplier **430**, and adder **432**; logic **434**, **436**, and **438**; and output value ResultOut **450**. In some embodiments, logic **434**, **436**, and **438** are AND gates. In some embodiments, additional signals are included as appropriate. In various embodiments, the computation unit of FIG. 4 is repeated for each of the plurality of computation units, such as computation unit **109**, of a computation array such as matrix processor **107** of FIG. 1. Computation unit **400** may be utilized to implement computational operations in parallel. In various embodiments, each computation unit of a computational array performs computations in parallel with the other computation units. In various embodiments, computation unit **400** is a sub-circuit of a matrix processor that includes the functionality for performing one or more multiply, add, accumulate, and/or shift operations. For example, computation unit **400** may be a sub-circuit that includes the functionality for performing a dot-product operation.

[0051] In some embodiments, Clock signal **410** is a clock signal received by computation unit **400**. In various embodiments, each computation unit of the computational array receives the same clock signal and the clock signal is utilized to synchronize the processing of each computation unit with the other computation units.

[0052] In the example shown, multiplier **430** receives and performs a multiplication operation on the input values data **404** and weight **402**. The output of multiplier **430** is fed to adder **432**. Adder **432** receives and performs an addition on the output of multiplier **430** and the output of logic **434**. The output of adder **432** is fed to accumulator **424**. In some embodiments, input values data **404** and weight **402** are lines that cross computation units and feed the corresponding data and/or weight to neighboring computation units. For example, in some embodiments, data **404** is fed to all computation units in the same column and weight **402** is fed to all computation units in the same row. In various embodiments, data **404** and weight **402** correspond to input elements fed to computation unit **400** from a data hardware data formatter and a weight hardware data formatter, respectively. In some embodiments, the data hardware data formatter and the weight hardware data formatter are data formatter **104** and weight formatter **106** of FIG. 1, respectively.

[0053] In some embodiments, ClearAcc signal **408** clears the contents of accumulator **424**. As an example, accumulation operations can be reset by clearing accumulator **424** and used to accumulate the result of multiplier **430**. In some embodiments, ClearAcc signal **408** is used to clear accumulator **424** for performing a new dot-product operation. For example, elements-wise multiplications are performed by multiplier **430** and the partial-dot-product results are added using adder **432** and accumulator **424**.

[0054] In various embodiments, accumulator **424** is an accumulator capable of accumulating the result of adder **432** and indirectly the result of multiplier **430**. For example, in some embodiments, accumulator **424** is configured to accumulate the result of multiplier **430** with the contents of accumulator **424** based on the status of ClearAcc signal **408**. As another example, based on the status of ClearAcc signal **408**, the current result stored in accumulator **424** may be ignored by adder **432**. In the example shown, accumulator **424** is a 32-bit wide accumulator. In various embodiments, accumulator **424** may be sized differently, e.g., 8-bits, 16-bits, 64-bits, etc., as appropriate. In various embodiments, each accumulator of the plurality of computation units of a computational array is the same size. In various embodiments, accumulator **424** may accumulate and save data, accumulate and clear data, or just clear data. In some embodiments, accumulator **424** may be implemented as an accumulation register. In some embodiments, accumulator **424** may include a set of arithmetic logic units (ALUs) that include registers.

[0055] In some embodiments, ResultEnable signal **412** is activated in response to a determination that data **404** is valid. For example, ResultEnable signal **412** may be enabled to enable processing

by a computation unit such as processing by multiplier **430** and adder **432** into accumulator **424**. [0056] In some embodiments, ResultCapture signal **414** is utilized to determine the functionality of multiplexer **426**. Multiplexer **426** receives as input ResultIn **406**, output of accumulator **424**, and ResultCapture signal **414**. In various embodiments, ResultCapture signal **414** is used to enable either ResultIn **406** or the output of accumulator **424** to pass through as the output of multiplexer **426**. In some embodiments, multiplexer **426** is implemented as an output register. In some embodiments, ResultIn **406** is connected to a computation unit in the same column as computation unit **400**. For example, the output of a neighboring computation unit is fed in as an input value ResultIn **406** to computation unit **400**. In some embodiments, the input of a neighboring computation unit is the computation unit's corresponding ResultOut value.

[0057] In some embodiments, shadow register **428** receives as input the output of multiplexer **426**. In some embodiments, shadow register **428** is configured to receive the output of accumulator **424** via multiplexer **426** depending on the value of ResultCapture signal **414**. In the example shown, the output of shadow register **428** is output value ResultOut **450**. In various embodiments, once a result is inserted into shadow register **428**, accumulator **424** may be used to commence new calculations. For example, once the final dot-product result is stored in shadow register **428**, accumulator **424** may be cleared and used to accumulate and store the partial result and eventually the final result of a new dot-product operation on new weight and data input values. In the example shown, shadow register **428** receives a signal ShiftEn signal **416**. In various embodiments, ShiftEn signal **416** is used to enable or disable the storing of values in the shadow register **428**. In some embodiments, ShiftEn signal **416** is used to shift the value stored in shadow register **428** to output value ResultOut **450**. For example, when ShiftEn signal **416** is enabled, the value stored in shadow register **428** is shifted out of shadow register **428** as output value ResultOut **450**. In some embodiments, ResultOut **450** is connected to a neighboring computation unit's input value ResultIn. In some embodiments, the last cell of a column of computation units is connected to the output of the computational array. In various embodiments, the output of the computational array feeds into a vector engine such as vector engine **111** of FIG. **1** for vector processing. For example, the output ResultOut **450** of a computation cell such as computation cell **109** of FIG. **1** may be fed into a processing element of a vector engine such as processing element **113** of vector engine **111** of FIG. **1**.

[0058] In the example shown, shadow register **428** is 32-bits wide. In various embodiments, shadow register **428** may be sized differently, e.g., 8-bits, 16-bits, 64-bits, etc., as appropriate. In various embodiments, each shadow register of the plurality of computation units of a computational array is the same size. In various embodiments, shadow register **428** is the same size as accumulator **424**. In various embodiments, the size of multiplexer **426** is based on the size of accumulator **424** and/or shadow register **428** (e.g., the same size or larger).

[0059] In some embodiments, logic **434**, **436**, and **438** receive signals, such as control signals, to enable and/or configure the functionality of computation unit **400**. In various embodiments, logic **434**, **436**, and **438** are implemented using AND gates and/or functionality corresponding to an AND gate. For example, as described above, logic **434** receives ClearAcc signal **408** and an input value corresponding to the value stored in accumulator **424**. Based on ClearAcc signal **408**, the output of logic **434** is determined and fed to adder **432**. As another example, logic **436** receives ResultEnable signal **412** and Clock signal **410**. Based on ResultEnable signal **412**, the output of logic **436** is determined and fed to accumulator **424**. As another example, logic **438** receives ShiftEn signal **416** and Clock signal **410**. Based on ShiftEn signal **416**, the output of logic **438** is determined and fed to shadow register **428**.

[0060] In various embodiments, computation units may perform a multiplication, an addition operation, and a shift operation at the same time, i.e., within a single cycle, thereby doubling the total number of operations that occur each cycle. In some embodiments, results are moved from multiplexer **426** to shadow register **428** in a single clock cycle, i.e., without the need of

intermediate execute and save operations. In various embodiments, the clock cycle is based on the signal received at Clock signal **410**.

[0061] In various embodiments, input values weight **402** and data **404** are 8-bit values. In some embodiments, weight **402** is a signed value and data **404** is unsigned. In various embodiments, weight **402** and data **404** may be signed or unsigned, as appropriate. In some embodiments, ResultIn **406** and ResultOut **450** are 32-bit values. In various embodiments ResultIn **406** and ResultOut **450** are implemented using a larger number of bits than input operands weight **402** and data **404**. By utilizing a large number of bits, the results of multiplying multiple pairs of weight **402** and data **404**, for example, to calculate a dot-product result, may be accumulated without overflowing the scalar result.

[0062] In some embodiments, computation unit **400** generates an intermediate and/or final computation result in accumulator **424**. The final computation result is then stored in shadow register **428** via multiplexer **426**. In some embodiments, multiplexer **426** functions as an output register and store the output of accumulator **424**. In various embodiments, the final computation result is the result of a convolution operation. For example, the final result at ResultOut **450** is the result of convolution between a filter received by computation unit **400** as input values using weight **402** and a two-dimensional region of sensor data received by computation unit **400** as input values using data **404**.

[0063] As an example, a convolution operation may be performed using computation unit **400** on a 2×2 data input matrix $\begin{bmatrix} d0 & d1 \\ d2 & d3 \end{bmatrix}$ corresponding to a region of sensor data and a filter corresponding to a 2×2 matrix of weights $\begin{bmatrix} w0 & w1 \\ w2 & w3 \end{bmatrix}$. The 2×2 data input matrix has a first row $[d0 \ d1]$ and a second row $[d2 \ d3]$. The filter matrix has a first row $[w0 \ w1]$ and a second row $[w2 \ w3]$. In various embodiments, computation unit **400** receives the data matrix via data **404** as a one-dimensional input vector $[d0 \ d1 \ d2 \ d3]$ one element per clock cycle and weight matrix via weight **402** as a one-dimensional input vector $[w0 \ w1 \ w2 \ w3]$ one element per clock cycle. Using computation unit **400**, the dot product of the two input vectors is performed to produce a scalar result at ResultOut **450**. For example, multiplier **430** is used to multiply each corresponding element of the input weight and data vectors and the results are stored and added to previous results in accumulator **424**. For example, the result of element $d0$ multiplied by element $w0$ (e.g., $d0 * w0$) is first stored in cleared accumulator **424**. Next, element $d1$ is multiplied by element $w1$ and added using adder **432** to the previous result stored in accumulator **424** (e.g., $d0 * w0$) to compute the equivalent of $d0 * w0 + d1 * w1$. Processing continues to the third pair of elements $d2$ and $w2$ to compute the equivalent of $d0 * w0 + d1 * w1 + d2 * w2$ at accumulator **424**. The last pair of elements is multiplied and the final result of the dot product is now stored in accumulator **424** (e.g., $d0 * w0 + d1 * w1 + d2 * w2 + d3 * w3$). The dot-product result is then copied to shadow register **428**. Once stored in shadow register **428**, a new dot-product operation may be initiated, for example, using a different region of sensor data. Based on ShiftEn signal **416**, the dot-product result stored in shadow register **428** is shifted out of shadow register **428** to ResultOut **450**. In various embodiments, the weight and data matrices may be different dimensions than the example above. For example, larger dimensions may be used.

[0064] In some embodiments, a bias parameter is introduced and added to the dot-product result using accumulator **424**. In some embodiments, the bias parameter is received as input at either weight **402** or data **404** along with a multiplication identity element as the other input value. The bias parameter is multiplied against the identity element to preserve the bias parameter and the multiplication result (e.g., the bias parameter) is added to the dot-product result using adder **432**. The addition result, a dot-product result offset by a bias value, is stored in accumulator **424** and later shifted out at ResultOut **450** using shadow register **428**. In some embodiments, a bias is introduced using a vector engine such as vector engine **111** of FIG. **1**.

[0065] FIG. **5** is a block diagram illustrating an embodiment of a cache-enabled microprocessor system for performing machine learning processing. The microprocessor system of FIG. **5** includes

hardware data formatters that interface with a cache to prepare input values for a computational array such as a matrix processor. In various embodiments, incorporating a memory cache and using hardware data formatters to populate the cache increases the throughput of the matrix processor and allows the microprocessor system to operate at a higher clock rate than would otherwise be allowed. In the example shown, microprocessor system **500** includes control unit **501**, memory **502**, cache **503**, data formatter **504**, weight formatter **506**, and matrix processor **507**. Input data and weight data are retrieved by hardware data formatters **504**, **506** from memory **502** via cache **503**. The retrieved input values are formatted using data formatter **504** and weight formatter **506** to prepare vector operands for matrix processor **507**. In some embodiments, data formatter **504** and weight formatter **506** include a logic circuit for preparing data for matrix processor **507** and/or a memory cache or buffer for storing and processing input data. For example, data formatter **504** may prepare N operands from a two-dimensional array retrieved from memory **502** via cache **503**. Weight formatter **506** may prepare M operands retrieved from memory **502** via cache **503** that correspond to weight values. Data formatter **504** and weight formatter **506** prepare the N and M operands to be processed by matrix processor **507**.

[0066] In various embodiments, microprocessor system **500** is microprocessor system **100** of FIG. **1** depicted with a memory and memory cache. With respect to microprocessor **100** of FIG. **1**, in various embodiments, control unit **501** is control unit **101**, data formatter **504** is data formatter **104**, weight formatter **506** is weight formatter **106**, and matrix processor **507** is matrix processor **107** of FIG. **1**. Further, with respect to microprocessor **100** of FIG. **1**, in various embodiments, data input **103** and weight input **105** of FIG. **1** are retrieved from memory **502** via cache **503**. In some embodiments, microprocessor system **500**, including at least hardware data formatter **504**, weight formatter **506**, and matrix processor **507**, performs the processes described with respect to FIGS. **7** and **8** and portions of processes described with respect to FIGS. **2** and **3**.

[0067] In some embodiments, matrix processor **507** is a computational array that includes a plurality of computation units. For example, a matrix processor receiving M operands and N operands from weight formatter **506** and data formatter **504**, respectively, includes $M \times N$ computation units. In the figure shown, the small squares inside matrix processor **507** depict that matrix processor **507** includes a logical two-dimensional array of computation units. Computation unit **509** is one of a plurality of computation units of matrix processor **507**. In some embodiments, each computation unit is configured to receive one operand from data formatter **504** and one operand from weight formatter **506**. Matrix processor **507** and computation unit **509** are described in further detail with respect to matrix processor **107** and computation unit **109**, respectively, of FIG. **1**. Input values to matrix processor **507** are received from data formatter **504** and weight formatter **506** and described in further detail with respect to inputs from data formatter **104** and weight formatter **106** to matrix processor **107** of FIG. **1**.

[0068] In the example shown, the dotted arrows between data formatter **504** and matrix processor **507** and between weight formatter **506** and matrix processor **507** depict a coupling between the respective pairs of components that are capable of sending multiple data elements such as a vector of data elements. In various embodiments, the data width of components data formatter **504**, weight formatter **506**, and matrix processor **507** are wide data widths and include the ability to transfer more than one operand in parallel. The data widths of components data formatter **504**, weight formatter **506**, and matrix processor **507** are described in further detail with respect to corresponding components data formatter **104**, weight formatter **106**, and matrix processor **107** of FIG. **1**.

[0069] In various embodiments, the arrows in FIG. **5** describe the direction data and/or control signals flow from component to component. In some embodiments, the connections depicted by the one-direction arrows in FIG. **5** (e.g., between data formatter **504** and cache **503**, between weight formatter **506** and cache **503**, and between cache **503** and memory **502**) may be bi-directional and thus the data and/or control signals may flow in both directions. For example, in some

embodiments, control signals, such as a read request and/or data, can flow from cache **503** to memory **502**.

[0070] In various embodiments, memory **502** is typically static random access memory (SRAM). In some embodiments, memory **502** has a single read port or a limited number of read ports. In some embodiments, the amount of memory **502** dedicated to storing data (e.g., sensor data, image data, etc.), weights (e.g., weight associated with image filters, etc.), and/or other data may be dynamically allocated. For example, memory **502** may be configured to partition more or less memory for data input compared to weight input based on a particular workload. In some embodiments, cache **503** includes one or more cache lines. For example, in some embodiments, cache **503** is a 1 KB cache that includes four cache lines where each cache line is 256 bytes. In various embodiments, the size of the cache may be larger or small, with fewer or more cache lines, have larger or smaller cache lines, and may be determined based on expected computation workload.

[0071] In various embodiments, hardware data formatters (e.g., data formatter **504** and weight formatter **506**) calculate memory addresses to retrieve input values from memory **502** and cache **503** for processing by matrix processor **507**. In some embodiments, data formatter **504** and/or weight formatter **506** stream data corresponding to a subset of values stored consecutively in memory **502** and/or cache **503**. Data formatter **504** and/or weight formatter **506** may retrieve one or more subsets of values stored consecutively in memory and prepare the data as input values for matrix processor **507**. In various embodiments, the one or more subsets of values are not themselves stored consecutively in memory with other subsets. In some embodiments, memory **502** contains a single read port. In some embodiments, memory **502** contains a limited number of read ports and the number of read ports is fewer than the data width of components data formatter **504**, weight formatter **506**, and matrix processor **507**. In some embodiments, hardware data formatters **504**, **506** will perform a cache check to determine whether a subset of values is in cache **503** prior to issuing a read request to memory **502**. In the event the subset of values is cached, hardware data formatters **504**, **506** will retrieve the data from cache **503**. In various embodiments, in the event of a cache miss, hardware data formatters **504**, **506** will retrieve the entire subset of values from memory **502** and populate a cache line of cache **503** with the retrieved values.

[0072] In some embodiments, control unit **501** initiates and synchronizes processing between components of microprocessor system **500**, including components memory **502**, data formatter **504**, weight formatter **506**, and matrix processor **507**. In some embodiments, control unit **501** coordinates access to memory **502** including the issuance of read requests. In some embodiments, control unit **501** interfaces with memory **502** to initiate read requests. In various embodiments, the read requests are initiated by hardware data formatters **504**, **506** via the control unit **501**. In various embodiments, control unit **501** synchronizes data that is fed to matrix processor **507** from data formatter **504** and weight formatter **506**. In some embodiments, control unit **501** synchronizes the data between different components of microprocessor system **500** including between data formatter **504**, weight formatter **506**, and matrix processor **507**, by utilizing processor specific memory, queue, and/or dequeue operations and/or control signals. Additional functionality performed by control unit **501** is described in further detail with respect to control unit **101** of FIG. **1**.

[0073] In some embodiments, microprocessor system **500** is utilized for performing convolution operations. For example, matrix processor **507** may be used to perform calculations, including dot-product operations, associated with one or more convolution layers of a convolution neural network. Data formatter **504** and weight formatter **506** may be utilized to prepare matrix and/or vector data in a format for processing by matrix processor **507**. Memory **502** may be utilized to store data such as one or more image channels captured by sensors (not shown). Memory **502** may also include weights, including weights in the context of convolution filters, determined by training a machine learning model for autonomous driving.

[0074] In various embodiments, microprocessor system **500** may include additional components

(not shown in FIG. 5), including processing components, such as a vector processor and a post-processing unit. An example of a vector processor and its associated functionality is vector engine **111** of FIG. 1. An example of a post-processing unit and its associated functionality is post-processing unit **115** of FIG. 1.

[0075] FIG. 6 is a block diagram illustrating an embodiment of a hardware data formatter, cache, and memory components of a microprocessor system. In the example shown, the components include memory **601**, cache **603**, and hardware data formatter **605**. Memory **601** is communicatively connected to cache **603** and cache **603** is communicatively connected to hardware data formatter **605**. Cache **603** includes four cache lines **611**, **613**, **615**, and **617**. Hardware data formatter **605** includes twelve read buffers **621-632**. Read buffers **621-632** are each 8-byte read buffers. In various embodiments, the number of and size of the read buffers may be fewer or more than depicted in the embodiment of FIG. 6. For example, read buffers **621-632** are sized to accommodate a **96** element input vector, where each element is 1-byte, to a computational array. In various embodiments, read buffers **621-632** may be implemented as a single wide register, a single memory storage location, individual registers, or individual memory storage locations, among other implementations, as appropriate. In some embodiments, memory **601** and cache **603** are memory **502** and cache **503** of FIG. 5, respectively. In some embodiments, hardware data formatter **605** is data formatter **104** and/or weight formatter **106** of FIG. 1. In some embodiments, hardware data formatter **605** is data formatter **504** and/or weight formatter **506** of FIG. 5.

[0076] In various embodiments, a control unit (not shown) such as control unit **101** of FIG. 1 and a computational array (not shown) such as matrix processor **107** of FIG. 1 are components of the microprocessor system. For example, a control unit sends signals to synchronize the processing of computational operations and/or access to memory **601**. In various embodiments, a computational array receives input vectors from one or more hardware data formatters as input operands. For example, a matrix processor may receive two vector inputs, one from a data formatter and one from a weight formatter, to perform matrix processing on. As another example, a matrix processor may receive two matrices, one from a data formatter and one from a weight formatter, to perform matrix processing on. In various embodiments, multiple clock cycles are needed to feed an entire matrix into a computational array. For example, in some embodiments, at most one row (and/or column) of a matrix is fed into a computational array each clock cycle.

[0077] In various embodiments, the output of hardware data formatter **605** is fed as input to a computational array such as matrix processor **107** of FIG. 1 and matrix processor **507** of FIG. 5. In various embodiments, each element of each read buffer of hardware data formatter **605** is fed into a computation unit of a computational array. For example, the first byte of read buffer **621** is fed into a first computation unit of a computational array, the second byte of read buffer **621** is fed into a second computation unit of a computational array, the third byte of read buffer **621** is fed into a third computation unit of a computational array, and so forth, with the last byte of read buffer **621** (i.e., the eighth byte) feeding into the eighth computation unit of a computational array. The next read buffer then feeds its elements into the next set of computation units. For example, the first byte of read buffer **622** is fed into a ninth computation unit of a computational array and the last byte of read buffer **632** is fed into a ninety-sixth computation unit of a computational array. In various embodiments, the size and number of the read buffers and the number of computation units may vary. As explained above, in the example shown, hardware data formatter **605** includes **12** read buffers **621-632** configured to each store eight consecutive bytes. Hardware data formatter **605** may be configured to feed into a computation unit that may receive at least one input vector of **96** 1-byte elements.

[0078] In some embodiments, only a portion of the elements in read buffers **621-632** is utilized as input to a computational array. For example, a two-dimensional **80**×**80** matrix may only utilize read buffers **621-630** (corresponding to **80** bytes, numbered bytes **0-79**) to feed an **80**-element row into a matrix processor. In various embodiments, hardware data formatter **605** may perform additional

processing on one or more elements of read buffers **621-632** to prepare the elements as input to a computational array. For example, a computational array may be configured to receive 48 16-bit elements instead of 96 8-bit elements and hardware data formatter **605** may be configured to combine pairs of 1-byte elements to form 16-bit elements to prepare a 48 16-bit input vector for the computational array.

[0079] In various embodiments, cache **603** is a memory cache of memory **601**. In some embodiments, memory **601** is implemented using static random access memory (SRAM). In some embodiments, cache **603** is a 1 KB memory cache and each cache line **611**, **613**, **615**, and **617** is 256 bytes. In various embodiments, reading data into cache **603** loads an entire cache line of data into one of cache lines **611**, **613**, **615**, and **617**. In various embodiments, cache **603** may be larger or small and have fewer or more cache lines. Moreover, in various embodiments, the cache lines may be a different size. The size and configuration of cache **603**, cache lines **611**, **613**, **615**, and **617**, and memory **601** may be sized as appropriate for the particular workload of computational operations. For example, the size and number of image filters used for convolution may dictate a larger or smaller cache line and a larger or smaller cache.

[0080] In the example shown, the dotted-lined arrows originating from read buffers **621-632** indicate whether the data requested by hardware data formatter **605** exists as a valid entry in cache **603** and in particular which cache line holds the data. For example, read buffers **621**, **622**, and **623** request data that is found in cache line **611**. Read buffers **626** and **627** request data that is found in cache line **613** and read buffers **630**, **631**, and **632** request data that is found in cache line **617**. In various embodiments, each read buffer stores a subset of values located consecutively in the memory. The subsets of values stored at read buffers **621**, **622**, and **623** may not be located consecutively in memory with the subsets of values stored at read buffers **626** and **627** and also may not be located consecutively in memory with the subsets of values stored at read buffers **630**, **631**, and **632**. In some scenarios, read buffers referencing the same cache line may store subsets of values that are not located consecutively in memory. For example, two read buffers may reference the same cache line of 256 bytes but different 8-byte subsets of consecutive values.

[0081] In the example shown, the data requested for read buffers **624**, **625**, **628**, and **629** are not found in cache **603** and are cache misses. In the example shown, an “X” depicts a cache miss. In various embodiments, cache misses must be resolved by issuing a read for the corresponding subset of data from memory **601**. In some embodiments, an entire cache line containing the requested subset of data is read from memory **601** and placed into a cache line of cache **603**. Various techniques for cache replacement may be utilized as appropriate. Examples of cache replacement policies for determining the cache line to use include First In First Out, Least Recently Used, etc.

[0082] In some embodiments, each of read buffers **621-632** stores a subset of values located consecutively in memory. For example, in the example shown, read buffer **621** is 8-bytes in size and stores a subset of 8-bytes of values stored consecutively in memory. In various embodiments, the values are located consecutively in memory **601** and read as a continuous block of values into a cache line of cache **603**. By implementing read buffers using the concept of a subset of values, where each of the values is located consecutively in memory, each read buffer is capable of loading multiple elements (e.g., up to eight elements for an 8-byte read buffer) together. In the example shown, a fewer number of reads are required than the number of elements to populate every read buffer with an element. For example, up to twelve reads are required to load 96-elements into the twelve read buffers **621-632**. In many scenarios, even fewer reads are necessary in the event that a cache contains the requested subset of data. Similarly, in some scenarios, a single cache line is capable of storing the data requested for multiple read buffers.

[0083] In some embodiments, read buffers **621-632** are utilized by hardware data formatter **605** to prepare input operands such as an vector of inputs for a computational array, such as matrix processor **107** of FIG. **1**. In some embodiments, the 96-bytes stored in read buffers **621-632** correspond to a 96-element input vector for a computational array. In some embodiments, hardware

data formatter **605** selects elements from read buffers **621-632** to accommodate a particular stride when performing a computational operation such as convolution. In some embodiments, hardware data formatter **605** selectively filters out the elements from read buffers **621-632** that are not required for the computational operation. For example, hardware data formatter may only utilize a portion of the elements from each read buffer (e.g., every other byte of a read buffer) as the input vector elements for the computational array. In some embodiments, the filtering is performed using a multiplexer to selectively include elements from read buffers **621-632** when preparing an input vector for a computational operation. In various embodiments, the unused bytes of the read buffer may be discarded.

[0084] As an example, in a scenario with a stride parameter set to two, the initial input elements for a convolution operation are every other element of a row of an input matrix. Depending on the input matrix size, the elements include the 1st, 3rd, 5th, and 7th elements, etc., for the first group of input elements necessary for a convolution operation. Read buffer **621** is configured to read the first 8 elements (1 through 8), and thus elements 2, 4, 6, and 8 are not needed for a stride of two. As another example, using a stride of five, four elements are skipped when determining the start of the next neighboring region. Depending on the size of the input data, the 1st, 6th, 11th, 16th, and 21st elements, etc., are the first input elements necessary for a convolution operation. The elements 2-5 and 7-8 are loaded into a read buffer **621** but are not used for calculating the first dot-product component result corresponding to each region and may be filtered out.

[0085] In various embodiments, each read buffer loads eight consecutive elements and can satisfy two elements for a stride of five. For example, read buffer **621** initiates a read at element 1 and also reads in element 6, read buffer **621** initiates a read at element 11 and also reads in element 16, read buffer **622** initiates a read at element 21 and also reads in element 26, etc. In some embodiments, the reads are aligned to multiples of the read buffer size. In some embodiments, only the first read buffer is aligned to a multiple of the read buffer size. In various embodiments, only the start of each matrix row must be aligned to a multiple of the read buffer size. Depending on the stride and the size of the input matrix, in various embodiments, only a subset of the read buffers may be utilized. In various embodiments, the elements corresponding to least twelve regions, one element for each read buffer **621-632**, are loaded and fed to a computational array in parallel. In various embodiments, the number of input elements provided in parallel to a computational array is at least the number of read buffers in the hardware data formatter.

[0086] In some embodiments, the elements not needed for the particular stride are filtered out and not passed to the computational array. In various embodiments, using, for example, a multiplexer, the input elements conforming to the stride are selected from the loaded read buffers and formatted into an input vector for a computational array. Once the input vector is formatted, hardware data formatter **605** feeds the input vector to the computational array. The unneeded elements may be discarded. In some embodiments, the unneeded elements may be utilized for the next dot-product component and a future clock cycle and are not discarded from read buffers **621-632**. In various embodiments, the elements not needed for implementing a particular stride are fed as inputs to a computational array and the computational array and/or post-processing will filter the results to remove them. For example, the elements not needed may be provided as input to a computation array but the computation units corresponding to the unnecessary elements may be disabled.

[0087] In some embodiments, hardware data formatter **605** formats the input vector for a computational array to include padding. For example, hardware data formatter **605** may insert padding using read buffers **621-632**. In various embodiments, one or more padding parameters may be described by a control unit using a control signal and/or instruction parameter.

[0088] In some embodiments, hardware data formatter **605** determines a set of addresses for preparing operands for a computational array. For example, hardware data formatter **605** calculates associated memory locations required to load a subset of values, determines whether the subset is cached, and potentially issues a read to memory for the subset in the event of a cache miss. In some

scenarios, a pending read may satisfy a cache miss. In various embodiments, hardware data formatter **605** only processes the memory address associated with the start element and end element of each read buffer **621-632**. In various embodiments, each read buffer **621-632** associates the validity of the cache entry for a subset of values with the memory addresses of the start and end values of the corresponding read buffer. In the example shown, read buffer **621** is configured to store 8-bytes corresponding to up to eight elements. In various embodiments, hardware data formatter **605** calculates the address of the first element and the address of the last element of read buffer **621**. Hardware data formatter **605** performs a cache check on the first and last element addresses. In the event either of the addresses is a cache miss, hardware data formatter **605** issues a memory read for 8-bytes starting at the address of the first element. In the event that both addresses are a cache hit from the same cache line, hardware data formatter **605** considers every element in the subset to be a valid cache hit and loads the subset of values from the cache via the appropriate cache line. In this manner, an entire row of elements may be loaded by processing the addresses of at most the first and last addresses of each read buffer **621-632** (e.g., at most 24 addresses).

[0089] FIG. 7 is a flow diagram illustrating an embodiment of a process for performing machine learning processing. The process of FIG. 7 describes a pipeline for slicing one or more matrices to fit a computational array, receiving a computational operation for the sliced matrix or matrices, preparing the data for performing the operation, and computing one or more results associated with the operation. Depending on the application, the process of FIG. 7 may be repeated on different slices of a matrix and the results combined. For example, a frame of image data larger than a computational array may be sliced into smaller matrices and computational operations performed on the sliced matrices. The results of multiple passes of FIG. 7 on different slices may be combined to generate the result of a computational operation on the entire frame. In various embodiments, the process of FIG. 7 is performed by a microprocessor system such as the microprocessor system of FIGS. 1 and 5. In various embodiments, the process of FIG. 7 is utilized to implement applications relying on computational operations such as convolution. For example, the process of FIG. 7 may be utilized to implement a machine learning application that performs inference using a machine learning model. In some embodiments, the process of FIG. 7 is utilized to implement the processes of FIGS. 2 and 3.

[0090] At **701**, one or more matrices may be sliced. In some embodiments, the size of a matrix, for example, a matrix representing a frame of vision data, is larger than will fit in a computational array. In the event the matrix exceeds the size of the computational array, the matrix is sliced into a smaller two-dimensional matrix with a size limited to the appropriate dimensions of the computational array. In some embodiments, the sliced matrix is a smaller matrix with addresses to elements referencing the original matrix. In various embodiments, the sliced matrix is serialized into a vector for processing. In some embodiments, each pass of the process of FIG. 7 may slice a matrix into a different slice and slices may overlap with previous slices. In various embodiments, a data matrix and a weight matrix may both be sliced, although typically only a data matrix will require slicing. In various embodiments, matrices may be sliced only at boundaries corresponding to multiples of the read buffer size of a hardware data formatter. For example, in the event each read buffer is 8-bytes in size, each row of a sliced matrix must begin with an address having a multiple of eight. In the event a matrix fits within the computational array, no slicing is required (i.e., the matrix slice used for the remaining steps of FIG. 7 is simply the original matrix). In various embodiments, the matrix slice(s) are used as input matrices for the computational operation of **703**.

[0091] At **703**, a computational operation is received. For example, a matrix operation is received by the microprocessor system. As one example, a computational operation requesting a convolution of an image with a filter is received. In some embodiments, the operation may include the necessary parameters to perform the computational operation including the operations involved and the operands. For example, the operation may include the size of the input operands (e.g., the size

of each input matrix), the start address of each input matrix, a stride parameter, a padding parameter, and/or matrix, vector, and/or post-processing commands. For example, a computational operation may describe an image data size (e.g., 96×96, 1920×1080, etc.) and bit depth (e.g., 8-bits, 16-bits, etc.) and a filter size and bit depth, etc. In some embodiments, the computational operation is received by a control unit such as control unit **101** of FIGS. **1** and **501** of FIG. **5**. In some embodiments, a control unit processes the computational operation and performs the necessary synchronization between components of the microprocessor system. In various embodiments, the computational operation is a hardware implementation using control signals. In some embodiments, the computational operation is implemented using one or more processor instructions.

[0092] At **705**, each hardware data formatter receives a data formatting operation. In some embodiments, the data formatting operation is utilized to prepare input arguments for a computational array such as matrix processor **107** of FIGS. **1** and **507** of FIG. **5**. For example, each hardware data formatter receives a data formatting operation that includes information necessary to retrieve the data associated with a computational operation (e.g., a start address of a matrix, a matrix size parameter, a stride parameter, a padding parameter, etc.) and to prepare the data to be fed as input into the computational array. In some embodiments, the data formatting operation is implemented using control signals. In some embodiments, the data formatting operation is received by a hardware data formatter such as data formatter **104** and **504** of FIGS. **1** and **5**, respectively, and weight formatter **106** and **506** of FIGS. **1** and **5**, respectively. In some embodiments, hardware data formatter is hardware data formatter **605** of FIG. **6**. In some embodiments, a control unit such as control unit **101** of FIGS. **1** and **501** of FIG. **5** interfaces with a hardware data formatter to process data formatting operations.

[0093] At **707**, data addresses are processed by one or more hardware data formatters. For example, addresses corresponding to elements of the computational operation are processed by one or more hardware data formatters based on the formatting operations received at **705**. In some embodiments, the addresses are processed in order for the hardware data formatter to load the elements (from a cache or memory) and prepare an input vector for a computational array. In various embodiments, a hardware data formatter first calculates a pair of memory addresses for each subset of values to determine whether a subset of elements exists in a cache before issuing a request to memory in the event of a cache miss. In various embodiments, a read request to memory incurs a large latency that may be minimized by reading elements from a cache. In some scenarios, all elements are read from a cache and thus require any cache misses to first populate the cache by issuing a read to memory. To minimize the latency for each read, in various embodiments, the reads are performed on subsets of elements (or values). In some embodiments, memory may only have a limited number of read ports, for example, a single read port, and all reads are processed one at a time. For example, performing 96 independent reads incurs the latency of 96 independent reads for a memory with a single read port. To reduce read latency, subsets of values are read together from memory into corresponding read buffers of a hardware data formatter. For example, using subsets of eight values, at most 12 memory reads are required to read 96 values. In the event some of the subsets are in the cache from previous memory reads, even fewer memory reads are required.

[0094] In various embodiments, subsets of values are prepared by determining the memory addresses for the start value of each subset (where each value corresponds to an element) and the end value of each subset. For example, to prepare a subset of 8-values each of 1-byte, a cache check is performed using the calculated address of the start value and the calculated address of the end value of the subset. In the event either of the addresses are cache misses, a memory read is issued to read 8-bytes from memory beginning at the address of the start value. In some embodiments, in addition to reading the requested 8-bytes from memory, an entire cache line of data (corresponding to multiple subsets) is read from memory and stored in the cache. In various embodiments, in the event the start and end addresses of a subset are cached at the same cache line,

the entire subset of values is considered cached and no cache check is needed for the remaining elements of the subset. The entire subset is considered cached in the event the start and end elements are cached in the same cache line. In various embodiments, the processing at **707** determines the addresses of the start value of the subset and the end value of the subset for each subset of values. In various embodiments, one read buffer exists for each subset of values. In various embodiments, read buffers of a hardware data formatter are read buffers **621-632** of hardware data formatter **605** of FIG. **6**.

[0095] In some embodiments, a stride parameter is implemented and non-consecutive subsets of values are loaded into each read buffer. In various embodiments, each subset of

[0096] continuous values includes one or more elements needed to implement a particular stride parameter. For example, for a stride of one, every value in a subset of values located consecutively in memory is a utilized element. As another example, for a stride of two, every other value located consecutively in memory is utilized and a subset of eight consecutive values includes four utilized elements and four that are not utilized. As another example, for a stride of five, a subset of eight values located consecutively in memory may include two utilized elements and six unused elements. For each subset of elements located consecutively in memory, the memory addresses for the start and end elements of the subset are determined and utilized to perform a cache check at **709**. In various embodiments, the start element of the subset is the first element of the subset. In some embodiments, the end element of the subset is the last element of the subset, regardless of whether the element is utilized to implement the stride parameter. In some embodiments, the end element of the subset is the last utilized element and not the last element of the subset.

[0097] In various embodiments, once the number of utilized elements that are included in a subset of consecutive elements is determined, the next subset of elements begins with the next element needed to satisfy the stride parameter. The next element may result in a memory location that is located at an address non-consecutive with the address of the last element of the previous subset. As an example, using a stride of five, four elements are skipped when determining the start of the next subset of values. Depending on the size of the input data, the 1st and 6th elements are stored in the first subset of values, 11th and 16th elements in the second subset of values, and 21st and 26th elements in the third subset of values, etc. In various embodiments, the second subset of values starts with the 11th element and the third subset of values starts with the 21st element. Each subset is located in memory at locations non-consecutive with the other subsets. Examples of unused elements in the first subset of values include the elements 2-5 and 7-8. In some embodiments, the first row of each matrix is aligned to a multiple of the subset size. In some embodiments, this alignment restriction is required to prevent gaps of invalid values between rows when a matrix is serialized. In some embodiments, all subsets are aligned to the multiple of the subset size.

[0098] In various embodiments, each subset of values is loaded in a read buffer such as read buffers **621-632** of FIG. **6**. Depending on the particular application (e.g., the stride, the size of the input matrix, the size of the read buffer, the number of read buffers, etc.), some of the read buffers of a hardware data formatter may not be utilized. In some scenarios, the number of input elements provided in parallel to a computational array is at least the number of subsets. For example, a hardware data formatter supporting twelve subsets of values can provide at least twelve elements in parallel to a computational array.

[0099] In some embodiments, the formatting performed by a hardware data formatter includes converting a matrix into a vector with elements of the vector fed to a computational array over multiple clock cycles. For example, in some embodiments, a matrix corresponding to data (e.g., image data) is formatted to prepare vectors corresponding to sub-regions of the data. In some embodiments, each element fed to a computational array for a particular clock cycle corresponds to the n-th element of a vector associated with a sub-region of the data. As an example, a 3×3 matrix may be formatted into a one-dimensional vector of nine elements. Each of the nine elements may be fed into the same computation unit of a computational array. In various embodiments, feeding

the 9 elements requires are least 9 clock cycles.

[0100] At **709**, a determination is made whether the data corresponding to the addresses determined for each subset at **707** are cached. For example, a cache check is performed on each subset by determining whether the data associated with the address of the start value of the subset and the address of the end value of the subset is in the same cache line. In various embodiments, a cache check is performed for each read buffer, such as read buffers **621-632** of FIG. **6**, of a hardware data formatter. In the event the data is cached, the processing continues to **713**. In various embodiments, the cache utilized is cache **503** of FIG. **5** and/or **603** of FIG. **6**. In the event the data is not cached, processing continues to **711**.

[0101] At **711**, each requested subset of data is read into the cache as an entire subset of values. In various embodiments, each subset data is read into the cache from memory. In some embodiments, the memory is memory **502** of FIGS. **5** and **601** of FIGS. **6**. In some embodiments, an entire cache line is read into the cache. For example, a cache miss for a subset of values results in loading the subset of values into a cache line along with the other data located consecutively with the subset of values in memory. In some scenarios, a single cache line is sufficient to cache multiple subsets.

[0102] At **713**, matrix processing is performed. For example, a matrix processor performs a matrix operation using the data cached and received by a hardware data formatter. In various embodiments, the cached data is received by the hardware data formatter and processed according to a formatting operation by a hardware data formatter into input values for matrix processing. In some embodiments, the processing by the hardware data formatter includes filtering out a portion of the received cached data. For example, in some embodiments, subsets of values located consecutively in memory are read into the cache and received by the hardware data formatter. In various embodiments, a computational operation may specify a stride and/or padding parameters. For example, to implement a specified stride for convolution, one or more data elements may be filtered from each subset of values. In some embodiments, only a subset of the elements from each of the subsets of values is selected to create an input vector for matrix processing.

[0103] In various embodiments, the matrix processor performs the computational operation specified at **703**. For example, a matrix processor such as matrix processor **107** of FIGS. **1** and **507** of FIG. **5** performs a matrix operation on input vectors received by hardware data formatters. In various embodiments, the matrix processor commences processing once all the input operands are made available. The output of matrix processing is fed to **715** for optional additional processing. In various embodiments, the result of matrix processing is shifted out of a computational array one vector at a time.

[0104] At **715**, vector and/or post-processing operations are performed. For example, vector processing may include the application of an activation function such as a rectified linear unit (ReLU) function. In some embodiments, vector processing includes scaling and/or normalization. In various embodiments, vector processing is performed on one vector of the output of a computational array at a time. In some embodiments, vector processing is performed by a vector processor such as vector engine **111** of FIG. **1**. In various embodiments, post-processing operations may be performed at **715**. For example, post-processing operations such as pooling may be performed using a post-processor unit. In some embodiments, post-processing is performed by a post-processing processor such as post-processing unit **115** of FIG. **1**. In some embodiments, vector and/or post-processing operations are optional operations.

[0105] FIG. **8** is a flow diagram illustrating an embodiment of a process for retrieving input operands for a computational array. The process of FIG. **8** describes a process for preparing data elements by a hardware data formatter for a computational array. For example, the input data is partitioned into subsets based on the number of read buffers of a hardware data formatter. The process of FIG. **8** is utilized to load the corresponding read buffers with data corresponding to subsets of values located consecutively in memory. By partitioning values into subsets based on memory location and performing a single read on the entire subset instead of an individual read for

each element, the latency incurred from accessing memory is reduced. In various embodiments, the process of FIG. 8 is performed by a microprocessor system such as the microprocessor system of FIGS. 1 and 5. In various embodiments, the process of FIG. 8 is implemented at 707, 709, 711, and 713 of FIG. 7. In various embodiments, the memory utilized by the process of FIG. 8 is memory 502 of FIGS. 5 and 601 of FIG. 6. In various embodiments, the cache utilized by the process of FIG. 8 is cache 503 of FIGS. 5 and 603 of FIG. 6. In various embodiments, the process of FIG. 8 is performed at least in part by a hardware data formatter such as the hardware data formatters of FIGS. 1, 5, and 6. For example, a hardware data formatter may be utilized to perform the steps of 801, 803, 805, 807, 809, 811, 813, and portions of 815. In some embodiments, the process of FIG. 8 is utilized to implement the processes of FIGS. 2 and 3.

[0106] In some embodiments, the process of FIG. 8 is performed in parallel on different read buffers and/or subset of values. For example, in a scenario with eight read buffers, the data to be loaded into the read buffers may be partitioned into at most eight subsets and the process of FIG. 8 is performed on each subset in parallel. In some embodiments, the number of subsets is based on capabilities of the cache and/or the memory. For example, the number of subsets may be based on how many simultaneous cache checks may be performed on the cache and/or the number of simultaneous reads to memory that may be issued.

[0107] At 801, the first subset of data elements located consecutively in memory is processed. In various embodiments, the first consecutive subset of data corresponds to the data element designated for the first read buffer of a hardware data formatter. In some embodiments, the address of the first element must be a multiple of the number of elements in each subset. For example, using an 8-byte read buffer, the address of the first element must be a multiple of eight.

[0108] At 803, start and end memory addresses are determined for the current subset. For example, the memory address of the start element of a subset and the memory address of the end element of a subset are determined. In various embodiments, the start and end addresses are determined by a hardware data formatter, such as the hardware data formatters of FIGS. 1, 5, and 6.

[0109] At 805, a determination is made on whether the subset of data is cached or pending a read. For example, a determination is made whether the data corresponding to the start and end addresses determined at 803 are cached at the same cache line or will be cached as a result of an already issued memory read. In some embodiments, a pending read for a different subset brings an entire cache line of data into memory and will result in caching the current subset. In the event the data is not cached or will not be cached as a result of a pending memory read, processing continues to 807. In the event the data is cached or will be cached by a pending memory read, processing continues to 811.

[0110] At 807, a determination is made on whether a memory read is already issued. In the event a memory read is already issued, processing completes for the current clock cycle. In the event a memory read has not been issued, processing continues to 809. In some embodiments, the memory is configured with a single read port (e.g., to increase density) and the memory can only process one read at a time. In various embodiments, the determination of whether a memory read has been issued is based on the capability of the memory configuration and/or the availability of memory read ports. Not shown in FIG. 8, in some embodiments, in the event an additional memory read is supported for the current clock cycle (despite a pending read), processing continues to 809; otherwise processing completes for the current clock cycle.

[0111] At 809, a read is issued to cache a subset of data elements. For example, a block of memory beginning at the start address determined at 803 and extending for the length based on the size of a read buffer is read from memory into the memory cache. In various embodiments, an entire cache line of memory is read into the memory cache. For example, in a scenario with a cache line of 256 bytes and read buffers each capable of storing 8-bytes, a memory read will read 256 bytes of continuous data into a cache line, which corresponds to 32 subsets of non-overlapping 8-byte values. In various embodiments, reading a subset of values as a single memory read request

reduces the latency associated with loading each element. Moreover, reading multiple subsets of values together may further reduce the latency by caching other subsets of values that may be associated with other read buffers. In some embodiments, loading multiple subsets of values takes advantage of potential locality between the subsets resulting in lower latency.

[0112] At **811**, a determination is made on whether there are additional subsets of data elements. In the event that every subset has been processed, processing continues to **813**. In the event that there are additional subsets to be processed, processing loops back to **803**. In some embodiments, depending on the input size, one or more read buffers of a hardware data formatter may not be utilized.

[0113] At **813**, a determination is made on whether all the data elements are cached. In the event some elements are not cached, processing completes for the current clock cycle to allow the non-cached data elements to be loaded from memory into the cache. In the event all the data elements are cached, the data elements are all available for processing and processing proceeds to **815**.

[0114] At **815**, matrix processing is performed. For example, the cached data elements are received at one or more hardware data formatters, formatted, and fed as input vector(s) to a computational array for processing. A computational array, such as matrix processor **107** of FIGS. **1** and **507** of FIG. **5**, performs matrix processing on the input vectors.

[0115] Although the foregoing embodiments have been described in some detail for purposes of clarity of understanding, the invention is not limited to the details provided. There are many alternative ways of implementing the invention. The disclosed embodiments are illustrative and not restrictive.

Claims

1. (canceled)
2. A microprocessor system, comprising: a computational array that includes a plurality of computation units, wherein an individual computation unit operates on a corresponding value of a group of values addressed from memory; and a hardware data formatter configured to (i) provide, based on a sampling parameter, a first portion of the group of values to the computational array for a first computation processing, and (ii) perform a cache check to obtain the first portion of the group of values, wherein the computational array disables particular computational units based on the sampling parameter.
3. The microprocessor system of claim 2, wherein the sampling parameter corresponds to an architecture of a machine learning model associated with the computational array.
4. The microprocessor system of claim 2, wherein the sampling parameter comprises a kernel size, a step size, a down-sampling factor, a stride, or a spatial extent associated with the computational array.
5. The microprocessor system of claim 2, wherein a computation unit of the plurality of computation units is configured to perform at least a dot-product component operation using the group of values in parallel.
6. The microprocessor system of claim 2, wherein an output of the computational array includes a scalar value.
7. The microprocessor system of claim 2, wherein a computation unit of the plurality of computation units includes an arithmetic logic unit, an accumulator, and a shadow register.
8. The microprocessor system of claim 2, wherein the group of values corresponds to an input channel of vision data or sensor data.
9. The microprocessor system of claim 8, wherein the sensor data is non-image sensor data.
10. The microprocessor system of claim 9, wherein the first portion is retrieved from a cache using a single cache read.
11. The microprocessor system of claim 2, wherein the group of values corresponds to a

convolution filter.

12. The microprocessor system of claim 11, wherein the convolution filter is constructed to identify features of an input data.

13. The microprocessor system of claim 2, wherein the hardware data formatter is further configured to (i) identify a second portion of the group of values not provided for the first computation processing, and (ii) provide the second portion of the group of values to the computational array for a second computation processing subsequent to the first computation processing.

14. The microprocessor system of claim 13, wherein the first portion of the group of values and the second portion of the group of values are retrieved from a single cache line.

15. The microprocessor system of claim 13, wherein the memory is configured to dynamically adjust an allocation between a first portion of the memory for a data input and a second portion of the memory for a weight input.

16. The microprocessor system of claim 13, wherein the hardware data formatter is configured to determine a corresponding start memory address for each of the first portion of the group of values and the second portion of the group of values.

17. The microprocessor system of claim 16, wherein the hardware data formatter is configured to determine a corresponding end memory address for each of the first portion of the group of values and the second portion of the group of values.

18. The microprocessor system of claim 16, wherein the cache check is performed for each of the first portion of the group of values and the second portion of the group of values based on determining whether a value stored at the determined starting memory addresses for the first portion of the group of values and the second portion of the group of values has been cached.

19. The microprocessor system of claim 2, wherein the cache check is performed for the first portion of the group of values based on determining whether a first value and a last value for the first portion of the group of values are stored in a cache.

20. The microprocessor system of claim 2, wherein the first portion of the group of values are synchronously provided together to the computational array to be processed in parallel in the first computation processing.

21. A method comprising: performing, by a hardware data formatter, a cache check to obtain a portion of a group of values associated with a computational operation; providing, based on a sampling parameter and by the hardware data formatter, the portion of the group of values to a computational array for a portion of computation processing of the computational operation; and disabling, based on the sampling parameter and by the computational array, particular computational units of the computational array when performing the portion of computation processing.
